Business Question and Visualization Report

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	5 Marks

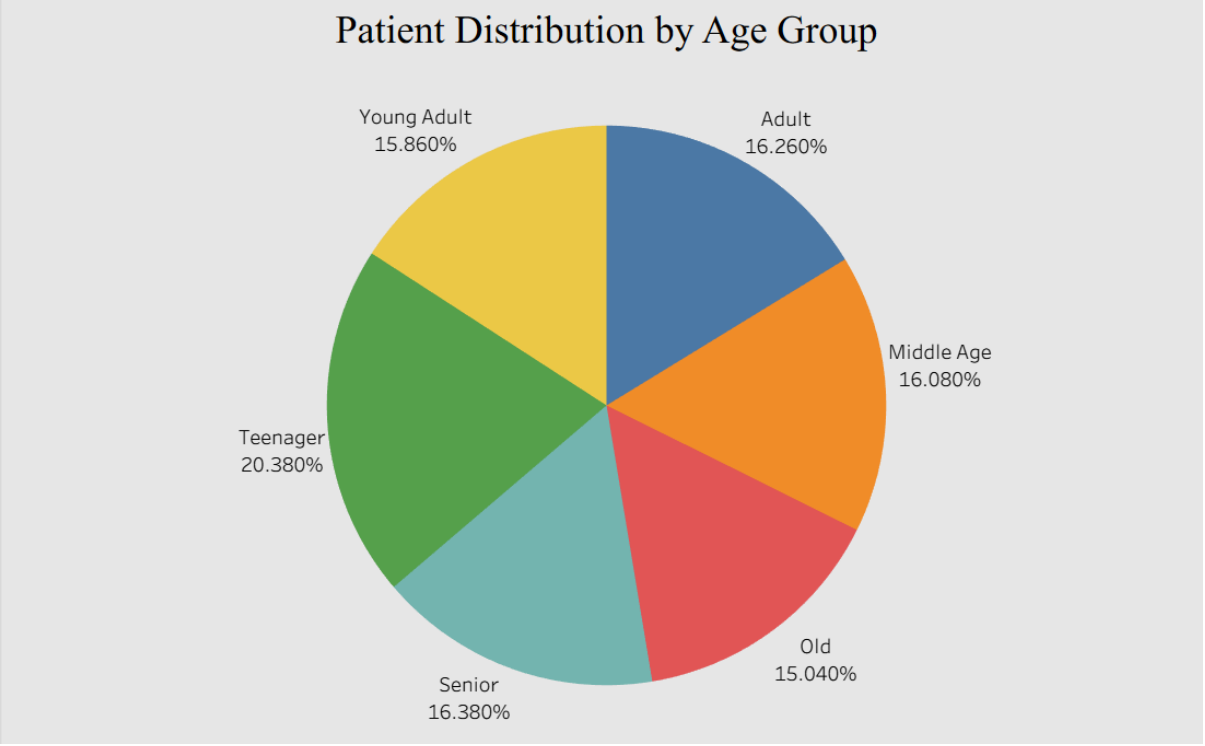
Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

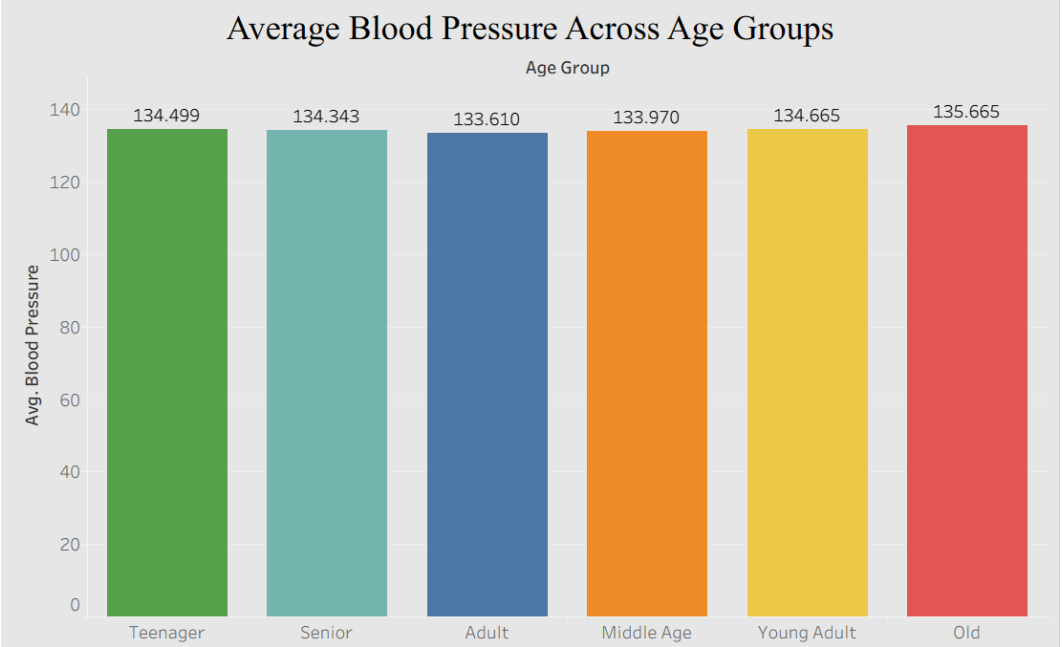
1. How are patients distributed across different age groups?

- Visualization: Bar chart showing the number of patients in each age group.
- Purpose: To understand the distribution of patients across age categories and identify groups with higher patient counts.
- Screenshot of visualisation:



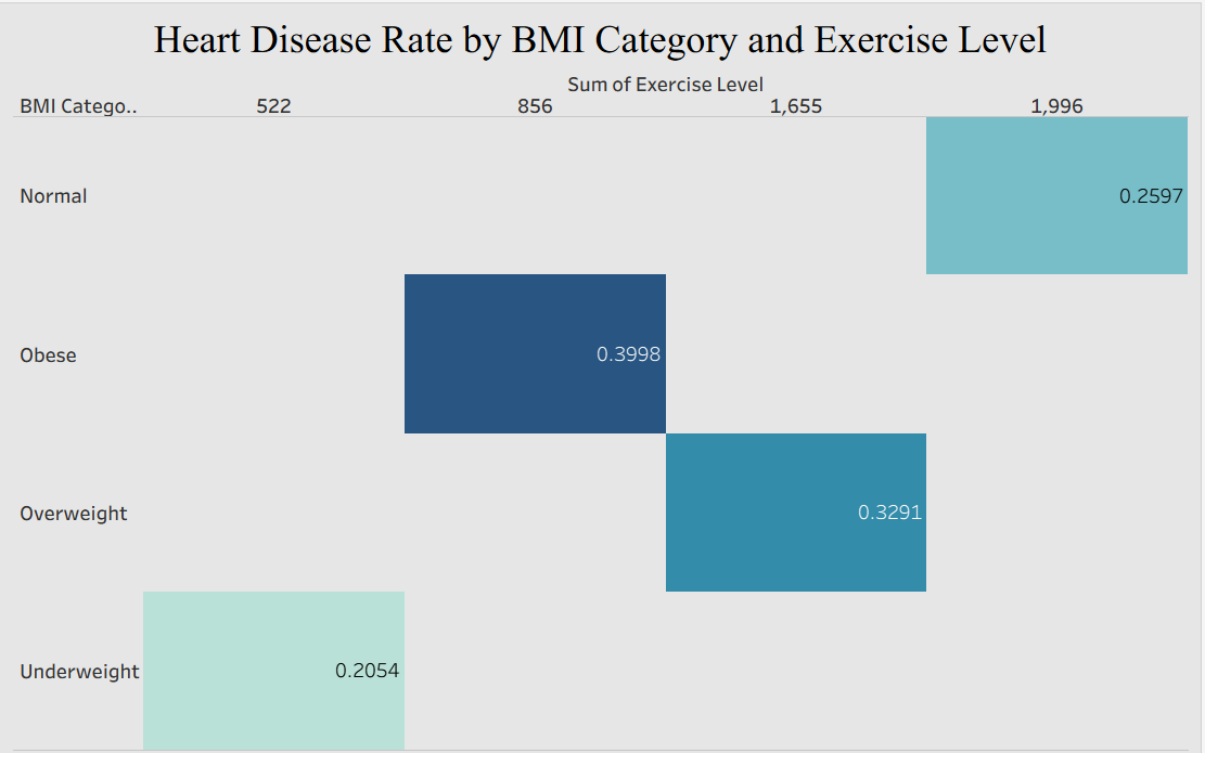
2. How does average blood pressure vary across age groups?

- Visualization: Bar chart showing average blood pressure for each age group.
- Purpose: To compare average blood pressure across different age categories.
- Screenshot of visualisation:



3. How does heart disease vary across BMI categories and exercise levels?

- Visualization: Heatmap comparing heart disease indicators across BMI categories and exercise levels.
- Purpose: To explore differences in the recorded heart disease indicators between BMI and exercise categories.
- Screenshot of visualisation: Insert your Tableau worksheet screenshot for Heart Disease by BMI Category and Exercise Level.



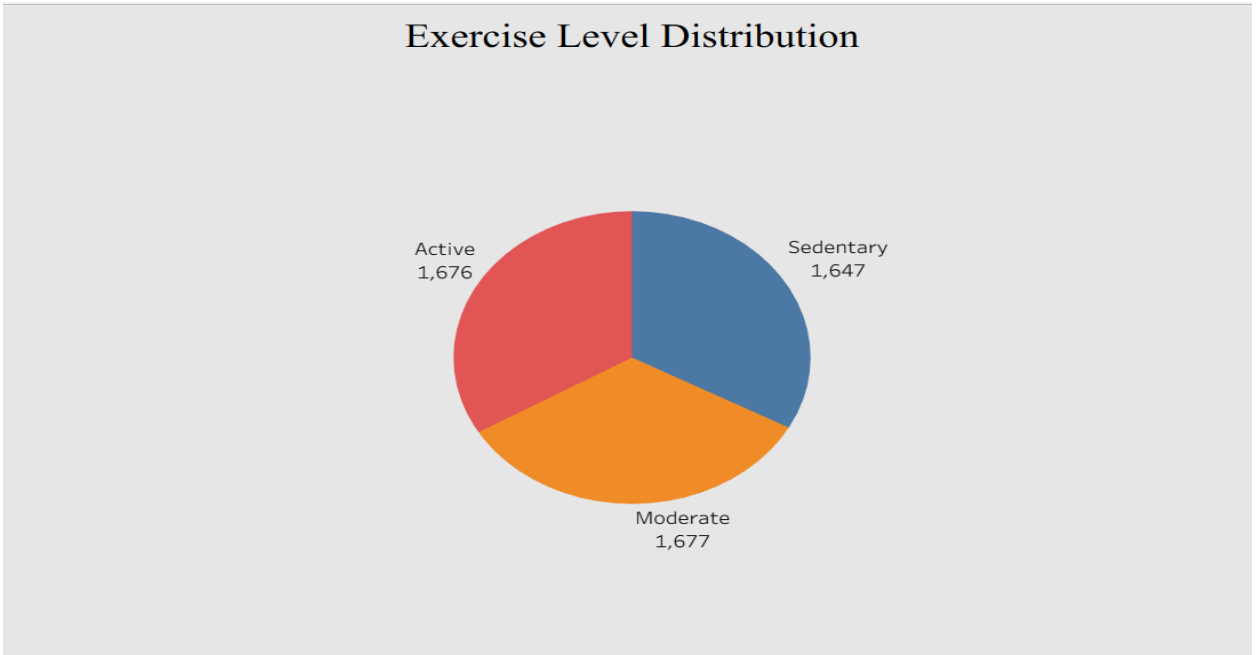
4. What is the overall number of patients and recorded disease cases?

- Visualization: KPI cards showing Total Patients, Diabetes Cases, Heart Disease Cases, and Stroke Cases.
- Purpose: To provide a quick overview of the patient dataset and the recorded disease indicators.
- Screenshot of visualisation:

Total Patients	Diabetes	Heart Disease	Stroke
5,000	1,746	1,499	1,243

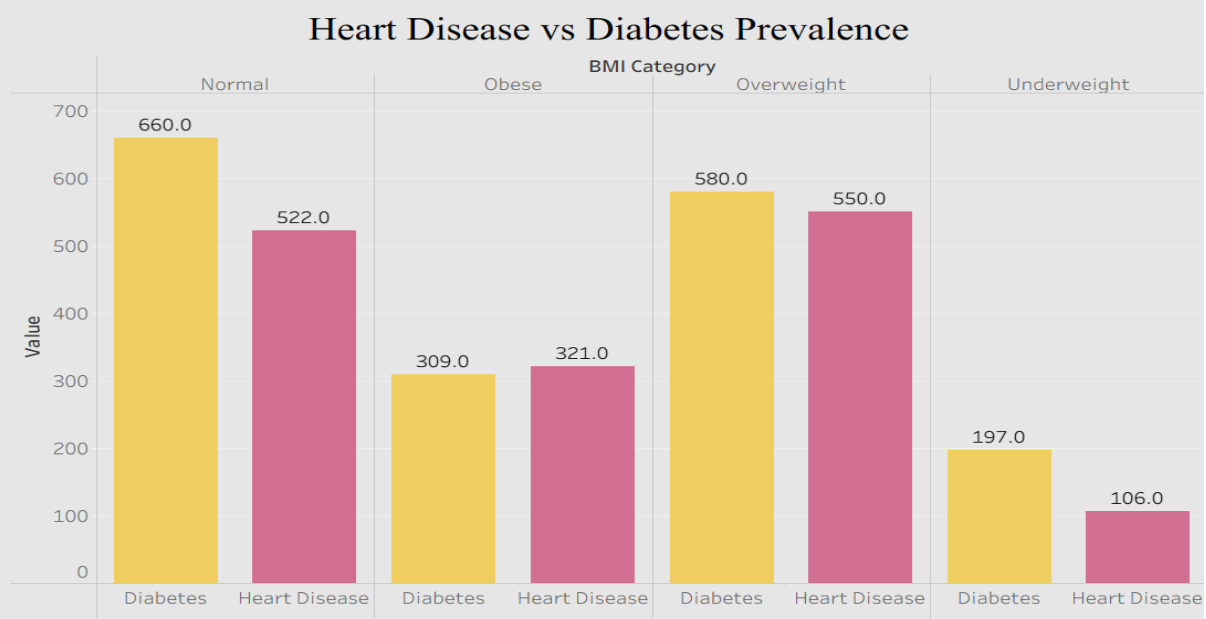
5. How does the distribution of exercise levels vary among patients?

- Visualization: Pie chart or donut chart showing the distribution of patients across Sedentary, Moderate, and Active exercise levels.
- Purpose: To understand patients' exercise habits and identify the most common activity level.
- Screenshot of visualisation:



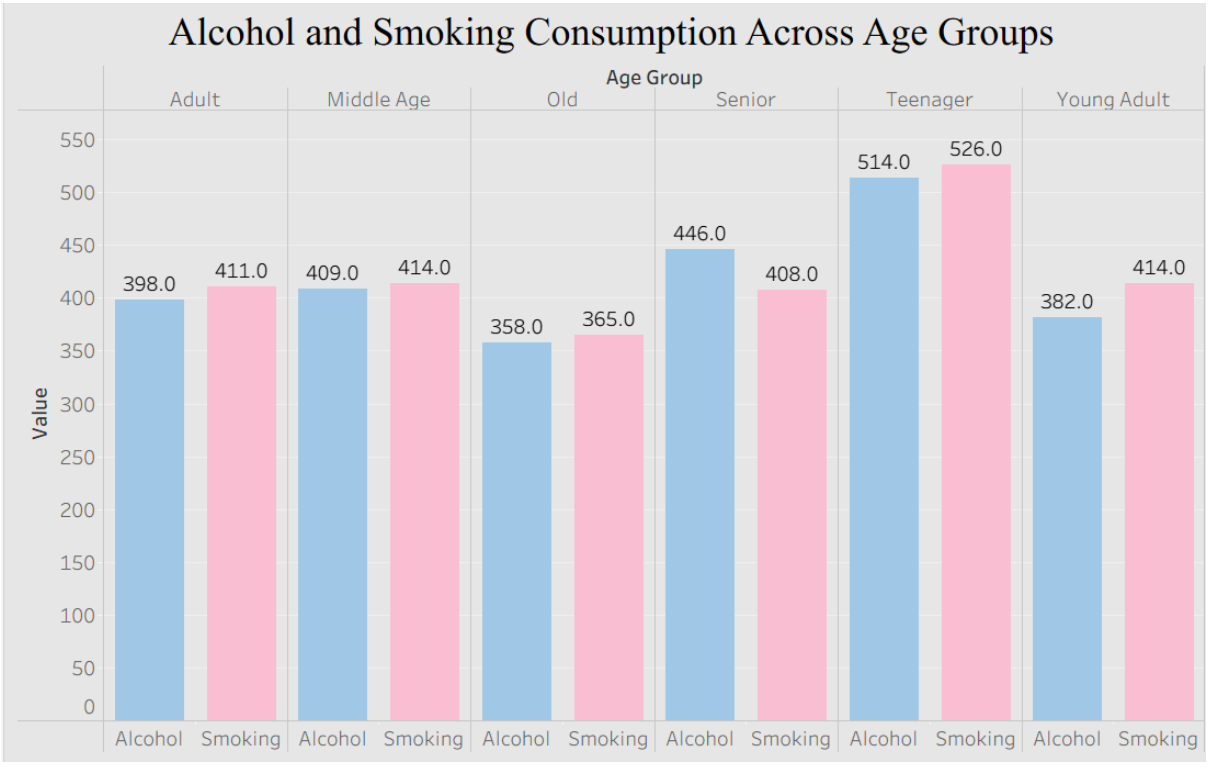
6. How do diabetes and heart disease cases compare across BMI categories?

- Visualization: Side-by-side bar chart comparing diabetes and heart disease cases across BMI categories.
- Purpose: To compare the recorded diabetes and heart disease indicators among patients in different BMI categories.
- Screenshot of visualisation:



7. How do alcohol consumption and smoking indicators vary across age groups?

- Visualization: Bar chart comparing alcohol and smoking indicators across different age groups.
- Purpose: To examine how the recorded alcohol and smoking indicators are distributed among different age categories.
- Screenshot of visualisation: Insert your Tableau worksheet screenshot for Alcohol and Smoking by Age Group.



8. How does fatigue vary across different age groups?

- Visualization: Bar chart showing the number of patients with fatigue in each age group.
- Purpose: To identify age groups with higher recorded fatigue counts and compare fatigue distribution across age categories.
- Screenshot of visualisation: Insert your Tableau worksheet screenshot for Fatigue by Age Group.

